

Date |15 March 2026
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Project Name |Credit Card Approval Prediction
Maximum Marks |4 Marks

Solution Requirements

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement	Requirement Description	Priority (High/Medium/Low)
1	Authentication	System features secure verification protocols including multi-factor authentication (MFA) for internal bank risk analysts and secure login access codes for external card applicants.	High
2	Authorization levels	Enforces clear role-based access levels: • Applicant: Restrictive access to see only their submission history and evaluation status. • Risk Analyst: Evaluation privileges to examine credit risks and authorize or override predictions. • Admin: System configuration and ML engine management.	High
3	External interfaces	Establishes real-time connection links via secure REST APIs with credit rating bureaus (e.g., CIBIL, Equifax) to securely stream credit history, and connects to core banking databases.	High
4	Transactions processing	Processes application data by passing financial details into the data transformation pipeline, runs automated machine learning inference, and logs approval decisions into the application database.	High

S.No	Requirement	Requirement Description	Priority (High/Medium/Low)
5	Reporting	Generates dynamic operational dashboards showing system approval metrics, demographic risk charts, model accuracy drifts, and provides data exports in Excel/PDF.	Medium
6	Business rules, etc.	Applies pre-screening business rules (e.g., age limit over 18, stable verified income streams). Flagged active bank bankruptcies invoke an immediate automated rejection before model evaluation.	High
7	Compliance to laws or regulations	Adheres to regional financial standards, fair lending guidelines, and rigorous data privacy acts. Strict masking and isolated storage apply to all personal identifier variables.	High
8	Other	Maintains an unalterable background transaction logging history capturing model input features and explainable feature ranking charts for compliance audits.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The end-to-end data processing and machine learning inference workflow must be highly optimized.	Model evaluation takes less than 500ms; UI forms load completely in less than 2 seconds.
2	Scalability	The web servers and prediction endpoints must dynamically scale during peak promotional marketing events.	Seamlessly accommodates up to 10,000 active concurrent connections.
3	Security & Data Privacy	All application data profiles, credit history files, and income statements must be robustly shielded.	AES-256 standards applied for inactive data; active data relies on TLS 1.3 encryption.
4	Reliability & Availability	The core predictive architecture must guarantee maximum operational continuity to eliminate transaction dropped states.	99.9% application service availability target measured monthly.
5	Usability & Accessibility	The internal application processing forms and tracking portals must remain intuitive and user-friendly.	The client-facing portal strictly obeys WCAG 2.1 Level AA accessibility models.
6	Other	The application codebase structure must support highly modular deployments so that updated ML weights can be rolled out.	Supports model weight rollouts or hot-swapping procedures within less than 5 minutes of downtime.